MINISTRY OF SCIENCE AND HIGHER EDUCATION OF THE REPUBLIC OF KAZAKHSTAN

INTERNATIONAL INFORMATION TECHNOLOGY UNIVERSITY

DEPARTMENT OF INFORMATION SYSTEM

PRACTICAL WORK #6

RESEARCH REPORT

Topic: Development of an Information System for Pharmacy Inventory and Medicine Management

Field of Study: Information Systems / Software Engineering

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1. Introduction

Modern pharmaceutical supply chains and internal pharmacy operations require high operational precision, reliable data integrity, and rapid stock verification. In traditional retail and hospital pharmacy settings, inventory control and medicine dispensing may be hindered by manual record-keeping, unindexed spreadsheets, and fragmented communication. These workflows can contribute to unexpected stockouts, financial losses caused by overlooked expiration dates, and dispensing errors that may affect patient safety. Specialized web-based information systems have therefore become an important technical and organizational option for improving pharmacy workflows.

A robust pharmacy inventory and medicine management system connects shelf-life rotation logic, relational database architecture, and responsive user interfaces. The literature reviewed for this report addresses inventory rotation, stock classification, demand forecasting, web architectures, database systems, software engineering, information security, and healthcare interoperability. Prior work cited in the supplied bibliography discusses pharmacy inventory controls and web-based inventory systems [1]–[10], while foundational texts cover software engineering, database concepts, and management information systems [11]–[15].

A research gap remains in empirical evaluations that compare transaction latency and operational error frequency between automated web systems and manual records in small-to-medium independent pharmacies. This report therefore focuses on the following research problem: how can a vulnerable manual inventory process be transitioned into a more efficient web-based information system, and how can its operational performance be evaluated quantitatively?

1.1 Research questions and hypotheses

1. How does implementing an automated web-based Pharmacy Inventory and Medicine Management System affect inventory-tracking speed, stockout rates, and order execution times compared with traditional manual or paper-based methods?

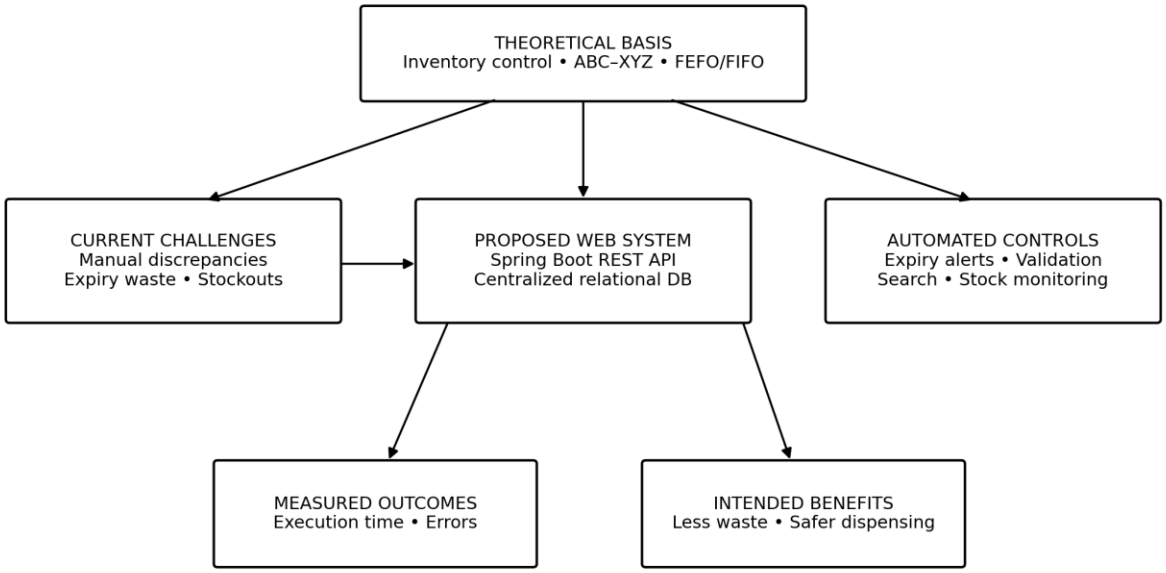
2. Is the reduction in operational errors and batch-verification latency statistically significant after deployment of a centralized database with automated stock alerts?

H_1 (working hypothesis): An automated pharmacy inventory system significantly reduces medicine-query and inventory-audit processing time ($p < 0.05$) and lowers operational error frequency.

H_0 (null hypothesis): There is no statistically significant difference in operational execution speed or error rates between the automated system and manual inventory tracking.

1.2 Significance and conceptual framework

Theoretically, the study contributes to discussion of software engineering in healthcare supply chains. Practically, the proposed architecture is intended to help pharmacies reduce avoidable inventory losses, improve stock visibility, and support more reliable workflows. These benefits remain intended outcomes until confirmed in a live operational environment.



Picture 1. Conceptual map linking inventory-control theory, current challenges, the proposed web solution, and intended outcomes.

The conceptual framework connects three layers: inventory-control principles (including ABC–XYZ classification and FEFO/FIFO rotation), the operational problems associated with manual records, and a proposed web architecture with centralized storage and automated alerts. Execution time and error observations are the immediate evaluation measures; reduced waste and improved patient safety are longer-term aims requiring further validation.

2. Methodology

A quantitative experimental comparison was described to evaluate the operational performance of the proposed pharmacy management system. The intended

measurements include task execution time, database query latency, HTTP transaction performance, and data-entry or expiry-monitoring errors.

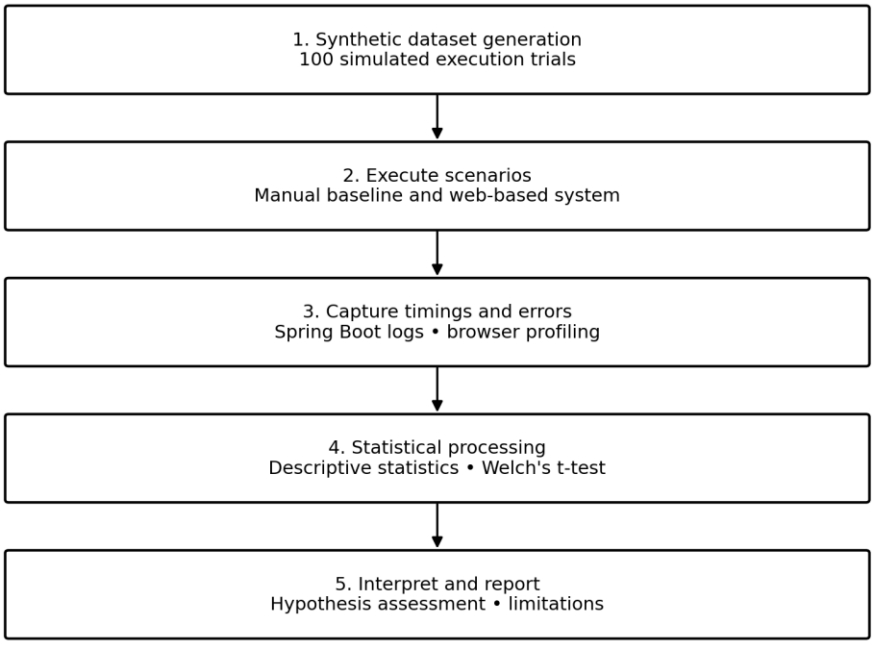
2.1 Population, sample, and conditions

The target population consisted of representative pharmacy workflows: medicine intake, batch-expiration tracking, active-ingredient searches, sales processing, and reorder triggering. The report describes 100 execution trials, split equally between a manual record-keeping baseline ($n_1 = 50$) and an automated web-based system ($n_2 = 50$). The trials are described as selected from a standardized test suite.

A sample size of 50 per group exceeds 30 observations, but this fact alone does not guarantee normality or validate a parametric test. The independence of observations, distribution of execution times, and variance assumptions should also be checked. Because raw trial-level data were not supplied with the report, those diagnostics cannot be independently verified here.

2.2 Instrumentation and procedure

The described instrumentation combined server-side performance logging in a Spring Boot application with browser profiling using Chrome DevTools. The logging plan covered SQL execution duration and connection-pooling latency at millisecond precision. A pilot of 10 transactions was described as a check of logging precision, database stability, and potential memory leaks. The dataset was described as synthetic and anonymized, using generated medicine names, random batch codes, and dummy supplier details.



Picture 2. Procedural workflow from synthetic data generation to statistical interpretation.

Phase	Operation	Tool / environment
1	Generate synthetic dataset (100 trials)	Python mock-data generator
2	Execute benchmark scenarios and log timings	Spring Boot middleware and database
3	Process descriptive and inferential statistics	Google Colab; pandas; scipy.stats
4	Assess hypotheses and document limitations	Researcher interpretation

Table 1. Summary of the described experimental workflow.

2.3 Statistical analysis

The planned descriptive statistics were mean, standard deviation, median, and range. An independent-samples t-test was used to compare execution time between the two groups. Since group variances differ substantially in the supplied summary (SD 14.8 seconds versus 2.9 seconds), Welch's t-test is the more appropriate default unless equal variances can be justified. The analysis is reproducible only when the trial-level data or sufficient summary statistics are available.

All evaluated records were described as synthetic, reducing the risk of exposing personal or proprietary information. No patient-level data were reported. A real

deployment would require appropriate access control, audit logging, data protection, and compliance with applicable pharmacy and health-data regulations.

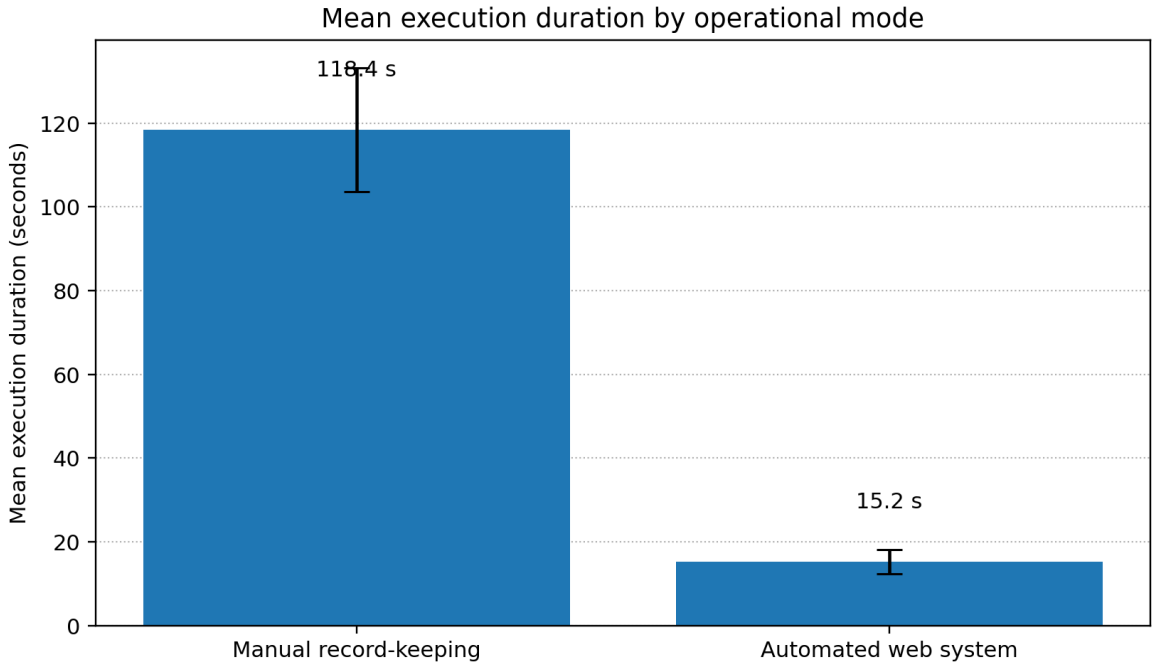
3. Results

The following results reproduce the summary values supplied in the report. Since the raw 100 trial observations were not included, the descriptive values can be presented, but the original inferential statistic cannot be reproduced exactly.

3.1 Descriptive statistics

Metric	Manual record-keeping	Automated web system
Sample size	50	50
Mean execution duration	118.4 s	15.2 s
Standard deviation	14.8 s	2.9 s
Median	119.0 s	15.0 s
Range	91.0–145.0 s	9.0–21.0 s
Reported error frequency	3.4 errors per batch	0 observed errors

Table 2. Descriptive comparison of manual and automated task execution.



Picture 3. Mean execution duration with standard-deviation error bars (summary statistics supplied in the report).

The automated group's mean duration was 103.2 seconds lower than the manual group's mean. This corresponds to an approximate 87.2% reduction in mean execution time, or a ratio of about 7.8:1 in favour of the automated workflow.

3.2 Inferential comparison

The supplied report stated $t = 50.4030$ and $p = 6.8726 \times 10^{-72}$. However, the reported t-statistic does not match the supplied group means, standard deviations, and sample sizes under a standard independent-samples calculation. Recalculation from the summary statistics using Welch's t-test gives $t \approx 48.38$, with approximately 50.3 degrees of freedom and $p < 0.001$. The exact p-value is extremely small; reporting $p < 0.001$ is sufficient and avoids implying more precision than the available data support.

Measure	Value	Interpretation
Difference in means	103.2 seconds	Manual mean minus automated mean
Relative reduction	≈87.2%	Reduction from 118.4 s to 15.2 s
Speed ratio	≈7.8×	118.4 s divided by 15.2 s
Welch t-statistic (recalculated)	≈48.38	Calculated from supplied summary statistics
Welch degrees of freedom	≈50.3	Approximate
p-value	< 0.001	Statistically significant under the stated model

Table 3. Recalculated comparison based on the reported summary statistics.

3.3 Operational error observations

The report records an average of 3.4 errors per batch under manual conditions and zero observed data-entry or expiry-oversight errors in the automated group. The number of batches, definition of an error, and method used to count errors were not specified in enough detail to calculate a formal significance test for this outcome. Therefore, the result is presented descriptively rather than as proof that the system eliminates errors in all circumstances.

4. Discussion

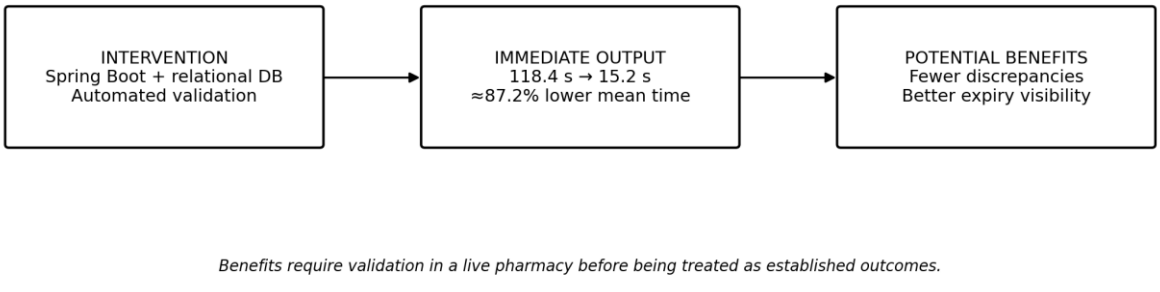
The supplied summary statistics indicate a large difference in mean execution time between the two tested conditions. The reduction from 118.4 seconds to 15.2 seconds is consistent with the expected operational advantage of searchable, centralized records over manual lookups. The recalculated Welch test indicates that this difference is statistically significant under the assumptions of the test. Statistical significance, however, does not by itself establish the same level of improvement in every pharmacy or workflow.

The findings are directionally consistent with the literature listed in the bibliography on pharmacy inventory management, web-based stock control, and software-supported workflows [1]–[10]. Centralized records, structured database queries, validation rules, and alerts can make stock information easier to retrieve and audit. The observed zero errors in the automated sample may reflect the controls built into the simulated workflow, but should not be generalized to real-world error elimination without a larger and independently audited pilot.

4.1 Limitations

- Simulation-based evidence: synthetic transactions do not fully reproduce real pharmacy operations, staff behaviour, medicine variability, or customer traffic.
- Missing raw observations: trial-level execution times and error records were not provided, preventing independent reproduction of all analyses and diagnostic checks.
- Unspecified operational details: hardware specifications, network conditions, task-by-task definitions, batch counts, and error definitions need clearer documentation.
- External validity: real pharmacies may experience variable internet bandwidth, concurrent cashier activity, different staff computer literacy, and integration constraints.
- Claims about downstream outcomes: reductions in expiration losses, stockouts, or clinical risk were proposed as benefits but were not directly measured in the reported experiment.

4.2 Implications and future work



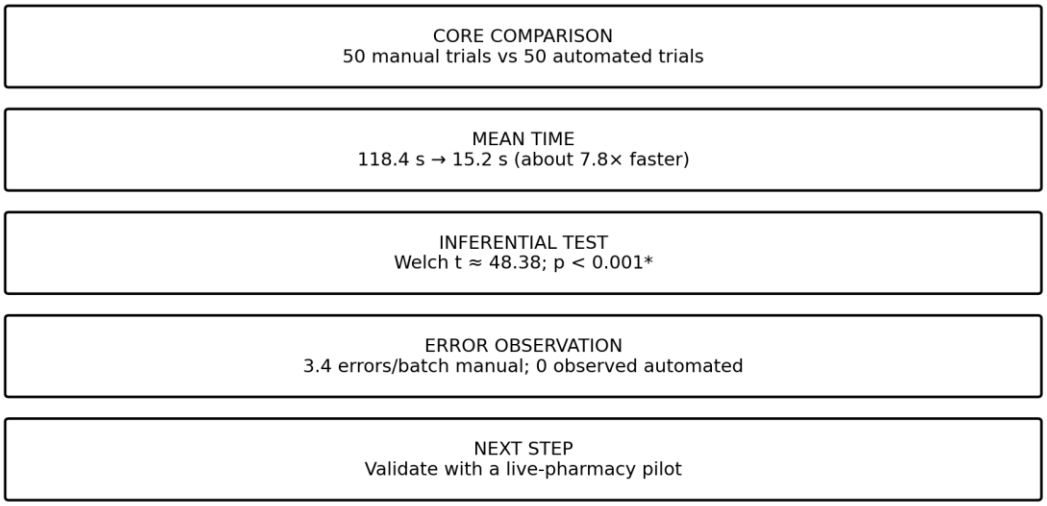
Picture 4. Logic model connecting system deployment, measured outputs, and potential longer-term implications.

Future work should deploy the system in a live pilot pharmacy and use a pre-registered protocol with defined tasks, error categories, and operational success criteria. It should also test barcode scanning, expiry notifications, backup and recovery, role-based access, and audit trails. Containerization or cloud-native deployment may improve maintainability and scaling, but scalability should be demonstrated through load testing rather than described as infinite.

5. Conclusion

This report presents a quantitative comparison framework for a web-based Pharmacy Inventory and Medicine Management System. In the supplied summary, the mean task duration decreased from 118.4 seconds under manual record-keeping to 15.2 seconds under the automated workflow, a reduction of approximately 87.2%. Recalculation from the reported summary statistics yields a Welch t-statistic of approximately 48.38 and $p < 0.001$. The report also records zero observed errors in the automated group, compared with an average of 3.4 errors per batch under manual conditions.

These results support the working hypothesis for the simulated execution-time comparison, subject to the assumptions and limitations described above. They do not establish that expiration losses or stockouts have been eliminated, nor do they prove readiness for unrestricted real-world deployment. Recommended next steps are staff training, consistent database design, security and backup testing, and a controlled live-pharmacy pilot with raw data collection and independent evaluation.



Picture 5. Executive summary of the reported comparison and recommended next step.

References

[1] E. Sukasih, G. Apriyanto, and A. Firdiansjah, “Drug inventory management in financial perspectives on pharmacy installations,” *IOSR Journal of Business and Management*, vol. 22, no. 8, ser. III, pp. 54–61, 2020, doi: 10.9790/487X-2208035461.

[2] A. Mustofa, K. Ekasari, and A. Kusmintarti, “Implementation of inventory accounting information systems at pharmacy (case study at Tirta Pharmacy),” *International Journal of Economics, Business and Accounting Research (IJEBAR)*, vol. 5, no. 2, pp. 153–160, Jun. 2021.

[3] C. A. Herlambang and J. Parung, “Information system design and inventory management on pharmacy business within ABC-XYZ analysis method,” *Airlangga Journal of Innovation Management*, vol. 2, no. 2, pp. 194–204, Oct. 2021.

[4] D. A. Efrilianda, K. Umam, and A. F. Aulia, “Inventory control and distribution of medicine stocks by using a just in time method based on interactive web applications,” *Journal of Physics: Conference Series*, vol. 1918, p. 042005, 2021, doi: 10.1088/1742-6596/1918/4/042005.

[5] G. J. Ken, H. Rajagopal, and S. S. Anjum, “Pharmacy warehouse management system,” in *Proceedings of the International Conference on Artificial Life and Robotics (ICAROB2023)*, Oita, Japan, Feb. 2023, pp. 663–668.

[6] M. Sumathi, I. Dhiraj, D. S. Mahita, and S. P. Raja, “Appointment booking and drug inventory system in healthcare services using blockchain technology,” *Advanced Distributed Computing and Artificial Intelligence Journal*, vol. 12, no. 1, article e31607, 2023, doi: 10.14201/adcaij.31607.

[7] A. L. Vasquez Ubaldo, J. A. Berrios Albines, J. L. Herrera Salazar, L. Andrade-Arenas, and M. Cabanillas-Carbonell, “Mobile application: A proposal for the inventory management of pharmaceutical industry companies,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 8, pp. 725–735, 2022.

[8] Y. Selvarathnam and M. A. Mohd Yunus, “MyIndah pharmacy inventory management system,” *Applied Information Technology and Computer Science*, vol. 6, no. 1, pp. 1923–1941, 2025, doi: 10.30880/aitcs.2025.06.01.107.

[9] H. Budianto, L. N. Angraeni, N. H. Nabila, and M. Fikri, “Development and quality evaluation of a web-based drug inventory system for pharmacy

management," *Nusantara Journal of Artificial Intelligence and Information Systems*, vol. 2, no. 1, pp. 27–38, Jun. 2026, doi: 10.47776/nuai.v2i1.1933.

[10] K. I. Latifa, L. Lazuardi, and C. Wiedyaningsih, "Hospital pharmacy information systems evaluation framework: A systematic review," *Generics: Journal of Research in Pharmacy*, vol. 6, no. 1, Jan. 2026.

[11] J. W. Creswell and J. D. Creswell, *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*, 5th ed. SAGE Publications, 2018.

[12] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hill Education, 2019.

[13] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2015.

[14] A. Silberschatz, H. F. Korth, and S. Sudarshan, *Database System Concepts*, 7th ed. McGraw-Hill, 2020.

[15] K. C. Laudon and J. P. Laudon, *Management Information Systems: Managing the Digital Firm*, 17th ed. Pearson, 2021.

Appendix A. Statistical analysis script template

The following template illustrates how the independent-groups comparison can be calculated if the 50 raw observations per condition are available. Replace the placeholder arrays with the actual trial-level timings before using the script. Summary statistics alone are not sufficient to reproduce all diagnostics.

```
import numpy as np
from scipy import stats

//Replace with the 50 raw execution times (seconds) for
each condition.
manual_times = np.array([...], dtype=float)
system_times = np.array([...], dtype=float)

print("Manual mean / SD:", manual_times.mean(),
manual_times.std(ddof=1))
print("System mean / SD:", system_times.mean(),
system_times.std(ddof=1))

//Welch's independent-samples t-test
result = stats.ttest_ind(manual_times, system_times,
```

```
equal_var=False)
print("Welch t-statistic:", result.statistic)
print("p-value:", result.pvalue)
```

Appendix B. Suggested relational schema

The original report refers to PostgreSQL relations named Medicines, Batches, Transactions, and Users but does not provide their full schema definitions. The following table proposes a high-level starting point; exact columns, constraints, and relationships should be finalized during implementation.

Entity	Example key fields	Purpose / relationship
Medicines	medicine_id (PK), name, active_ingredient, dosage_form, strength, barcode	Master record for each medicine; one medicine may have multiple batches.
Batches	batch_id (PK), medicine_id (FK), batch_code, expiry_date, quantity, supplier_id	Tracks lot-specific quantity and expiry; links to Medicines.
Transactions	transaction_id (PK), batch_id (FK), user_id (FK), transaction_type, quantity, timestamp	Records receipts, sales, adjustments, and stock movements.
Users	user_id (PK), username, password_hash, role, status	Supports authentication and role-based access; passwords must be stored as secure hashes.

Implementation note: enforce foreign keys, non-negative stock constraints, transaction auditing, least-privilege access, secure password hashing, backups, and appropriate retention rules. The schema above is a conceptual proposal, not a verified description of an implemented database.